Instantaneous Zero-Error U.F.O. Detection with Nullary Neural Networks

Ian F.V.G. Hunter
Commutable distance to UFO hot-spot — Newgrange, Meath, Ireland[*]

Abstract

In this newly declassified, and mostly uncensored, paper, we present our work on A.L.I.E.N.S. — A series of further and further optimal Neural Networks for the instantaneous classification of Unidentified Flying Objects (U.F.Os) & Unidentified Aerial Phenomena (U.A.Ps).

Not only is the performance of our software instantaneous, our software achieves 100% accuracy in all image and video benchmarks. It is resistant to the wear and tear of age and cannot be forcibly removed. In the spirit of advancing humanity, we have already installed this revolutionary software on all devices, digital or analogue, camera-capable or not. You can even use this paper. Just point it at something.

1 Introduction

Unidentified Flying Objects (U.F.Os) are of significant importance to government and military groups worldwide. There have been many instances of coverups and governmental conspiracies throughout modern [4][5][10]. Of course, the U.S. government is still recovering from [redacted]. While there are many skeptical of the field, we are confident that in the event of an alien invasion, the lack of research funding in this area will be regretted.

In this paper we favor the use of 'U.F.O.' in this publication over the newer 'U.A.P.' as U.A.P. has been used to navigate the field away from the connotations of extraterrestrial life[15], but our research is explicitly concerned with these sightings.

Our research in this paper is important for a great many reasons. To list them all here would require an entire secondary paper. But to give readers a flavor of what this work enables, consider that an optimal classification algorithm such as ours allows us to point at UFOs in the sky with zero error. This could mean we can target missile systems, laser beams, and uh, messages of greetings too I guess.

2 A.L.I.E.N.S.

A.L.I.E.N.S. stands for **A**dvanced **L**ocational **I**ntelligence for **E**xtraterrestial **N**avigational **S**ystems. It also spells out Aliens.

2.1 Version 1 — YOLO: Your Orbit Location Observed

Because Artificial Intelligence is the solution to all problems[1], our research began with a re-implementation of one of the state of the art Neural Networks — the YOLO object detection network [12]. We did not have any images of confirmed UFOs in our dataset, so artificial images were created as a training set, including photographs of far-away weather balloons, lenticular clouds, commercial aircraft with blurred company logos, lens flares, photography under heavy influence, and those LEDs on your electronics that glow annoyingly at night.

We very quickly reached a 100% classification rate on our data set, thanks to the unique property of U.F.Os to become I.F.Os (Identified Flying Objects) when classified. Because of this phenomenon, anything classified under the algorithm will, correctly, be classified correctly as negative. Whether the target is flying or not is not particularly relevant as many U.F.O.s do not even fly. They are many references to objects as U.F.O.s even if they are hovering or stationary in the air, landed or crashed. [1] [14] [3] [9] [8]. Of course, we cannot hope to understand the engineering limits of beings of superior intellect. We have provided Figure 1 to help you understand our classification system.

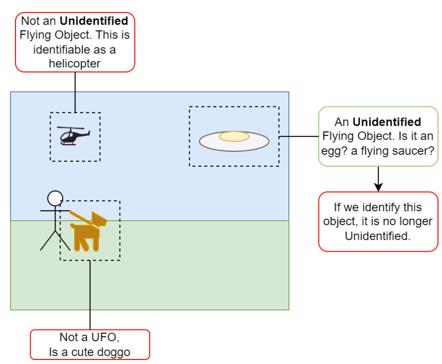

Figure 1: Classification of U.F.O.s with A.L.I.E.N.S.. All cases can be classified as NOT UFO.

2.2 Version 2 — FTL: Faster Than Last version

We were eager to reduce the computation required to get this result. If aliens were to invade, we need to mobilize the entire world, not just those with GPU or NPUs.

With lengthy analysis, we optimized A.L.I.E.N.S. over several properties:

- Sparse Computation[13] — Increasing the number of zero computations in our network
- Quantization down to binary weights and activations via BinaryConnect[2]
- Input resolution reduction
- Channel and Dimension Pruning[6] allowing us to remove compute from the network.

The results of our aggressive optimization resulted in the network seen in Figure 2. An ultra-minimal network

[*] The neolithic site of Newgrange is one of the top 3 places to spot a U.F.O. in Ireland[7]

[1] See: Company Reports for Investors over the past few years

consisting of a single input, hidden and output node, with a single binarized sparse weight. We also removed the bounding box prediction and replaced it with a more fundamental 'U.F.O' or 'not U.F.O'[2]

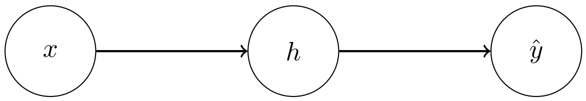

Figure 2: The Neurons within A.L.I.E.N.S. v2. The hidden layer's incoming weight value is 0. We have optimized away Bias values.

3 Version 3 (Final) — I-BLVE: Inference-free Binary-free Locator of Visual Entities

Despite having the fastest, smallest and most optimial Neural Network known to man, we were still not satisfied. The final and ultimate iteration of our A.L.I.E.N.S. algorithm culminated in the creation of I-BLVE, an algorithm that transcends traditional electronic boundaries.

After months of exclusive training and tens of thousands of our research department's budget, we came to a stunning conclusion — If our network parameters were converging so consistently towards zero — We could forgo the tradition of binary and represent our network with null values exclusively.

Because A.L.I.E.N.S. consists of exclusively null values, which do not alter the output, The I-BLVE variant can operate on any hardware, software or physical matter. All that matters is that the default conversion from a null output or "No Result" is interpreted by downstream software as a zero or "NOT UFO" state. See an example integration in 3. As you can see, the I-BLVE function requires no input and the output is not modified in the function body, so it can be easily substituted for any other operation that does not return a result - like petting a dog.

```
#include <stdbool.h>
#include <iostream>

void I_BLVE(bool * _){}

int main(){
    bool result = false;
    I_BLVE(&result);
    if(result){
        std::cout << "UFO Detected" << std::endl;
    } else {
        std::cout << "UFO Not Detected" << std::endl;
    }
}
```

Figure 3: A C++ example of I_BLVE integration

Thus, our algorithm has reached its ultimate form.

[2] That is the question. Well, actually, I suppose it's the answers.

4 Conclusion

In this paper we have detailed our A.L.I.E.N.S. algorithm and its evolution. This novel algorithm is the first full implementation of Nullary Neural Network use (as far as the authors are aware). Our I-BLVE variant demonstrates that optimizing to omission, disregarding traditional limitations like "bytes" and "cycles" can lead to infinitely better performance over traditional methods.

The future is bright for non-silicon based compute and we expect more development from our fellow researchers now that this work has been brought to light.

4.1 Other Related Work

We are not the only researchers to come across such revolutionary developments. LenPeg is a super-optimal image compression algorithm with similar performance[11].

4.2 Future Work

As mentioned within the paper, we expect this to be of use in high security military applications including the protection of Earth for the upcoming [redacted].

While we wait for that event, we have identified some other domains suitable for our Nullary Neural Network approach and their default null values that must be set beforehand:

- 'Good Boy/Good Girl' Dog Detector - I-BLVE(True).
- Presence of microplastics in food - I-BLVE(True).
- Relationship advice (Trained on Reddit Posts) - I-BLVE("DIVORCE").

Note that these use-cases will have undefined behaviour when not pointed at their intended target (e.g. a car may not be a Good Boy). We hope new innovations can address this limitation.

Finally, we would like to see future researchers investigating into the optimization of Large Language Models to use Zero-Hot encoding, under the hypothesis that the 'Silence is Golden' rule many of us were taught in our youth, may apply to AI models just as well as it does to school-children.

4.3 Acknowledgments

We would like to thank [redacted], [redacted], [redacted] — Particularly for use of their [redacted] facility. [redacted] for security clearance, [redacted] for clearing up the goo (Sorry!). The governments of [redacted], [redacted], [redacted] and [redacted].

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